

Problem 7.14

In this section the present value is assumed to be the value of an investment at time 0. We can determine the present value of an investment at any time n not necessarily at time 0.

A payment of \$10 is to be made at time 7 years.

- (a) Determine the present value of this payment at time 4 years if the annual compound interest is 6%.
- (b) Determine the present value of this payment at time 4 years if the annual simple interest is 6%.

Solution

The payment of \$10 occurs at time 7, so its value at time 4 is found by discounting it back 3 years.

(a) Compound interest at 6%

Under annual compound interest,

$$\text{Value at time 4} = 10(1.06)^{-3} = \frac{10}{(1.06)^3} \approx 8.3962.$$

Therefore, the present value at time 4 is

$$\boxed{\$8.3962}.$$

(b) Simple interest at 6%

Under annual simple interest,

$$10 = X(1 + it),$$

with $i = 0.06$ and $t = 3$. Thus,

$$10 = X(1 + 0.06 \cdot 3) = 1.18X.$$

So,

$$X = \frac{10}{1.18} \approx 8.4746.$$

Therefore, the present value at time 4 is

$$\boxed{\$8.4746}.$$